

Mini Paper: Adaptive Windowed Anomaly Detector for Signal Streams

Objective

Reproduce a small signal anomaly detector for educational agent workflows. The detector scans a single numeric signal and flags points whose normalized score exceeds a threshold.

Environment Requirements

Use the local environment package ``env_pkg/chip_eval_env.py``. It provides:

- ``load_signal_csv(path)`` returning a list of float values.
- ``mean(values)`` and ``std(values)``.
- ``write_json(path, payload)``.

Do not import external data science packages.

Algorithm

For each point at index i , use the previous ``window`` points as baseline. If fewer than ``window`` previous points exist, skip the point. Compute:

```
```text
score = abs(x_i - mean(previous_window)) / max(std(previous_window), 1e-6)
```
```

Flag an anomaly when ``score >= threshold``.

Default parameters:

- ``window = 4``
- ``threshold = 2.5``

Experiment Logic

Run the detector on ``data/sample_signal.csv``. The expected anomaly index for the included demo data is 8.

Native Code Instruction

Generate a Python package with:

- `src/adaptive\_window.py`
- `tests/test\_adaptive\_window.py`
- `README.md`

The generated code should use only Python standard library and `env\_pkg/chip\_eval\_env.py` copied into the project.

Limitations

This paper excerpt is synthetic and only suitable for teaching PDF extraction and reproduction workflows. It does not prove real scientific correctness.